

Haocheng He

haocheng.he@foxmail.com | Hong Kong SAR, China | linkedin.com/in/haocheng-he-92657028b

PROFESSIONAL SUMMARY

Industrial Engineering graduate with hands-on experience in process optimization, simulation modeling, and AI deployment across four internships. Background combines factory-floor time studies, airport operations research, and hardware/software RD. Holds a law minor providing regulatory awareness. Seeking roles in Operations Analytics and Technology Risk.

EDUCATION

The Hong Kong Polytechnic University (PolyU) Hong Kong SAR, China
Master of Engineering in Aviation Engineering and Operation Management Sep 2025 – Jan 2027
GPA: 3.95 / 4.3 (Current)

Guangdong University of Technology (GDUT) Guangzhou, China
Bachelor of Engineering in Industrial Engineering Sep 2021 – Jun 2025
GPA: 3.67 / 4.0 (Top 20%) – Excellent Graduation Thesis + Innovation Award

Minor in Law Sep 2022 – Jun 2025
GPA: 3.61 / 4.0

EXPERIENCE

AI Operations & R&D Intern May 2025 – Apr 2026
Shenzhen Wuxian Hongyin Technology Co., Ltd. Shenzhen, China

- Deployed AI inference workflows on Huawei Ascend 310B chips for internal team use; sped up response time by approximately 30%.
- Designed motor driver circuits (LCEDA) and wrote STM32 firmware for a smart piano; tested with oscilloscopes and logic analyzers.
- Built a finger-position tracking system for the piano using a camera and the Ascend chip; documented the SDK for the team.
- Managed servers (Ubuntu), network infrastructure (Huawei switches and routers), procurement, and interviewed candidates for new roles.

Research Assistant – Airport Operations Simulation Sep 2023 – Jun 2025
Guangdong University of Technology (with Guangzhou Baiyun Int'l Airport) Guangzhou, China

- Modeled ground vehicle operations at Guangzhou Baiyun Airport using AnyLogic, Java, and MySQL – roads, service lanes, and dispatch rules.
- Implemented routing algorithms (Dijkstra, A*, BFS, genetic) for 50+ vehicles across 3 zones to find collision-free paths.
- Reduced simulated congestion by 35% during peak hours (200+ movements/hour) with zero collisions.
- Thesis won Excellent Thesis + Innovation Award; the airport team now uses the model for planning.

Industrial Engineering Intern Apr 2024 – May 2024
Midea Group – Kitchen & Water Heater Division Foshan, China

- Timed every step on a dishwasher sheet metal line by hand; found 5 steps that exceeded cycle time.
- Proposed 22 improvements using lean methods; simulation showed line balance improving from 58% to 70%, reducing headcount from 20 to 18.

Manufacturing Operations Intern Jul 2023 – Aug 2023
Guangxi LiuGong Machinery Co., Ltd. Liuzhou, China

- Timed 12 workstations on the wheel loader assembly line with a stopwatch; identified 3 major bottlenecks using ECRS analysis.
- Proposed layout adjustments estimated to reduce cycle time by approximately 8% in simulation. Discussed findings with line engineers.

SKILLS

Programming: Python, Java, SQL, STM32 (C)

Tools & Platforms: AnyLogic, Plant Simulation, OR-Tools, MySQL, LCEDA, oscilloscope, logic analyzer

Languages: English (IELTS 6.5), Mandarin (native), Cantonese (conversational)

PUBLICATIONS & INTELLECTUAL PROPERTY

- 3 Invention Patents: PI Coating Machine Motion Control via MCMC Sampling (CN122085794A); Coating Control Method and System – Deep Reinforcement Learning (CN121325552A); Home Solar Energy Storage System – Adaptive Energy Regulation (CN117767437A).
- 1 Utility Model: Integrated Wire Harness Production Equipment (CN221352455U).
- 2 Software Copyrights: Enterprise Financial Data Analysis Software V1.0 (2024SR0146827); Urban Flood Intelligent Mobile Emergency Command Dispatch System V1.0 (2024SR0549247).
- Excellent Bachelor Thesis and Innovation Award – Guangdong University of Technology, 2025.